3.3 Quadratic Functions

Many relationships encountered in mathematics, science, engineering, economics, and data analysis involve a variable raised to the power of two. Such relationships are described by **quadratic equations**.

Definition

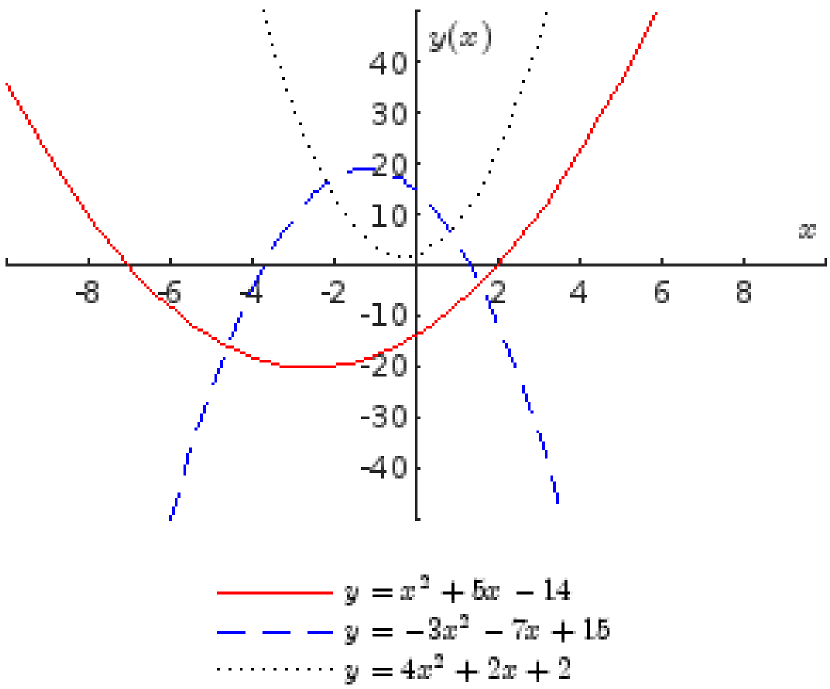
The **quadratic equation** is an equation that can be written in the form

$$y = ax^2 + bx + c,$$

where a, b, and c are constants. $a \neq 0$ and x is the independent variable.

3.3.1 The Graph of a Quadratic Function

The graph of a quadratic equation is called a **parabola**.



Definition

A parabola is a smooth curve whose shape and orientation depend on the coefficients of a, b and c.

- What is the orientation?
  - If $a > 0$, the parabola opens upwards.
  - If $a < 0$, the parabola opens downwards.
- Is it broad or narrow compared to $y = x^2$?
  - $|a| > 1$: narrower.
  - $|a| < 1$: broader.
- The turning point of the curve is called the **vertex**. Where this is depends on the coefficients:
  - Positive c will push it up.
  - Negative c will push it down.
  - Positive or negative b will push it down if $a > 0$ (up if $a < 0$).
  - $b = 0$: on the y-axis.
  - $b > 0$: left of the y-axis if $a > 0$ (right if $a < 0$).
  - $b < 0$: right of the y-axis if $a > 0$ (left if $a < 0$).

The constant c is the y-intercept, as in the linear case (if $x = 0$ then the equation becomes $y = a \times 0^2 + b \times 0 + c = c$).

The curve can cross the x-axis (at $y = 0$) twice, once (just touching it) or never. If it does cross the x-axis, we can calculate the values of x where this occurs by solving:

$$ax^2 + bx + c = 0$$

The **solutions** of $0 = ax^2 + bx + c$ are the same as the x-intercepts of $y = ax^2 + bx + c$ and are also known as the **roots** of $ax^2 + bx + c$.

Definition

The roots of a quadratic equation are the x-coordinates of the points where the corresponding parabola intersects the x-axis.

3.3.2 Solving by Factorisation

Some quadratic equations can be solved by expressing the quadratic expression as a product of two factors.

Worked Example

Solve

$$x^2 + 5x + 6 = 0$$

We seek two numbers whose product is 6 and whose sum is 5.

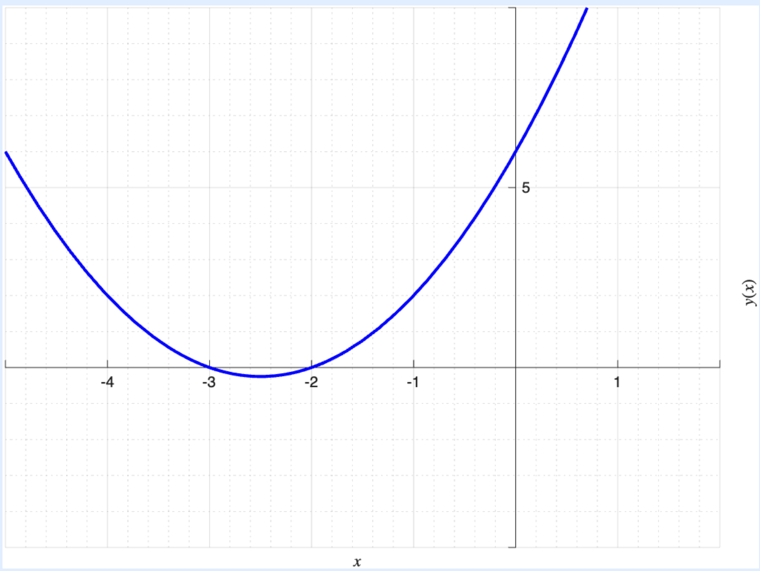
$$x^2 + 5x + 6 =$$

Therefore,

A product is zero if at least one factor is zero. Hence

Therefore,

The graph looks as follows:



Worked Example

Solve

$$x^2 - x - 20 = 0$$

We seek two numbers whose product is -20 and whose sum is -1.

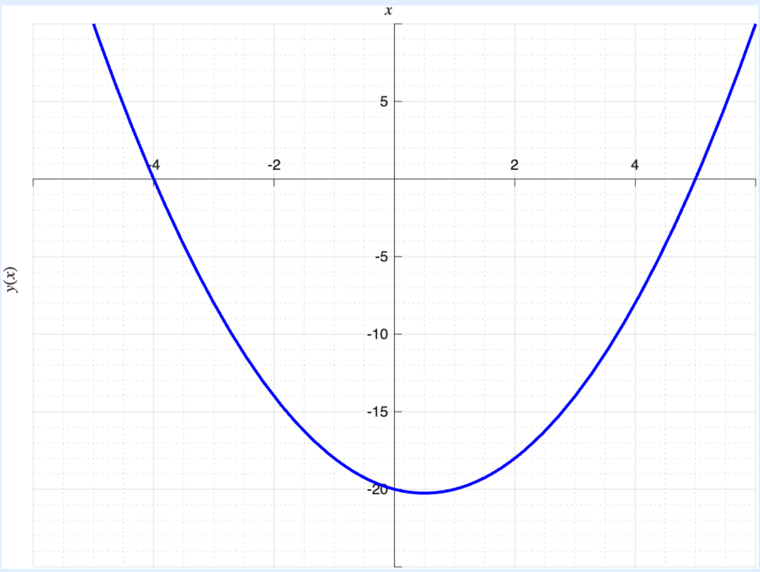
$$x^2 - x - 20 =$$

Therefore,

A product is zero if at least one factor is zero. Hence

Therefore,

The graph looks as follows:



3.3.3 The Quadratic Formula

Not all quadratic equations factorise easily. The quadratic formula can be used to solve any quadratic equation.

Definition

Given

$$ax^2 + bx + c = 0$$

the solutions are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Worked Example

Solve

$$2x^2 + 3x - 2 = 0$$

Here, $a =$, $b =$ and $c =$. Substituting into the quadratic formula gives

Hence,

Worked Example

Solve

$$x^2 - 11x + 28 = 0$$

Here, $a =$, $b =$ and $c =$. Substituting into the quadratic formula gives

Hence,

3.3.3.1 Factorisation from the Solutions

If there are two distinct, real roots x_1 and x_2 to the quadratic $y = ax^2 + bx + c$, then it is possible to re-write the quadratic in the fractorised form:

$$y = a(x - x_1)(x - x_2)$$

However, if there are real and repeated roots, $x_1 = x_2$, then it is possible to re-write the quadratic in the form:

$$y = a(x - x_1)^2$$

3.3.4 The Discriminant

The expression under the square root sign in the quadratic formula has special significance.

Definition

For the quadratic equation $ax^2 + bx + c = 0$, the quantity

$$\Delta = b^2 - 4ac$$

is called the **discriminant**. The discriminant determines the number and type of solutions.

- If $\Delta > 0$, the equation has two distinct real roots.
- If $\Delta = 0$, the equation has one repeated real root.
- If $\Delta < 0$, the equation has no real roots.

Definition

The number of real roots corresponds to the number of intersections between the parabola and the x-axis.

Graph Behaviour	Nature of Roots
Crosses the x-axis twice	Two distinct real roots
Touches the x-axis once	One repeated real root
Does not meet the x-axis	No real roots, i.e. complex roots

3.3.5 Applications of Quadratic Equations

Quadratic models occur naturally in many practical situations.

Engineering Application

Projectile Motion

Ignoring air resistance, the height of a projectile can often be modelled by

$$h(t) = at^2 + bt + c,$$

where t represents time. To determine when the projectile reaches the ground, we solve

$$h(t) = 0$$

This requires finding the roots of a quadratic equation.

Engineering Application

Revenue and Profit Models

Quadratic functions frequently arise in economic and business models. For example,

$$P(x) = -2x^2 + 80x - 300$$

may represent profit as a function of production level x. Analysing the graph allows us to identify the production level that maximises profit.

3.3.6 Exercises

Exercise

1. Solve the following equations by factorisation where possible.

   (a) $x^2 + 7x + 12 = 0$
   (b) $x^2 - 9x + 20 = 0$
   (c) $x^2 + 8x + 7 = 0$
   (d) $x^2 + 4x - 5 = 0$

2. Use the quadratic formula to solve:

   (a) $2x^2 + 7x + 3 = 0$
   (b) $3x^2 - 2x - 8 = 0$

3. Determine the nature of the roots using the discriminant.

   (a) $x^2 + 2x + 5 = 0$
   (b) $x^2 - 6x + 9 = 0$
   (c) $2x^2 + x - 4 = 0$

4. A ball is projected vertically and its height is modelled by

$$h(t) = -5t^2 + 20t + 25$$

   (a) Determine the time when the ball reaches the ground.
   (b) Sketch the graph.

3.3 Solutions to Exercise 3.3.6

1. Solve the following quadratic equations by factorisation.

   (a) $x^2+7x+12=0$. We require two numbers whose product is 12 and whose sum is 7. These numbers are 3 and 4.

   $$x^2+7x+12=0$$
   $$(x+3)(x+4)=0$$

   Therefore,

   $$x+3=0 \quad \text{or} \quad x+4=0$$
   $$x=-3 \qquad\qquad x=-4$$

   Therefore, the solutions are $x=-3$ and $x=-4$.

   (b) $x^2-9x+20=0$. We require two numbers whose product is 20 and whose sum is -9. These numbers are -4 and -5.

   $$x^2-9x+20=0$$
   $$(x-4)(x-5)=0$$

   Therefore,

   $$x-4=0 \quad \text{or} \quad x-5=0$$
   $$x=4 \qquad\qquad x=5$$

   Therefore, the solutions are $x=4$ and $x=5$.

   (c) $x^2+8x+7=0$. We require two numbers whose product is 7 and whose sum is 8. These numbers are 1 and 7.

   $$x^2+8x+7=0$$
   $$(x+1)(x+7)=0$$

   Therefore,

   $$x+1=0 \quad \text{or} \quad x+7=0$$
   $$x=-1 \qquad\qquad x=-7$$

   Therefore, the solutions are $x=-1$ and $x=-7$.

(d) $x^2 + 4x - 5 = 0$. We require two numbers whose product is -5 and whose sum is 4. These numbers are -1 and 5.

$$x^2 + 4x - 5 = 0$$
$$(x-1)(x+5) = 0$$

Therefore,

$$x - 1 = 0 \qquad \text{or} \qquad x + 5 = 0$$
$$x = 1 \qquad\qquad x = -5$$

Therefore, the solutions are $x = 1$ and $x = -5$.

2. (a) Solve the quadratic equation $2x^2 + 7x + 3 = 0$ using the quadratic formula. Here, $a = 2$, $b = 7$ and $c = 3$. Substituting into the formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
$$x = \frac{-7 \pm \sqrt{7^2 - 4(2)(3)}}{2(2)}$$
$$= \frac{-7 \pm \sqrt{25}}{4}$$
$$= \frac{-7 \pm 5}{4}$$

Therefore,

$$x = \frac{-7-5}{4}$$
$$= \frac{-12}{4}$$
$$= -3$$

or

$$x = \frac{-7+5}{4}$$
$$= \frac{-2}{4}$$
$$= -\frac{1}{2}$$

Therefore, the solutions are $x = -3$ and $x = -\frac{1}{2}$.

(b) Solve the quadratic equation $3x^2 - 2x - 8 = 0$ using the quadratic formula. Here, $a = 3$, $b = -2$ and $c = -8$. Substituting into the formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\begin{aligned} x &= \frac{- -2 \pm \sqrt{(-2)^2 - 4(3)(-8)}}{2(3)} \\ &= \frac{2 \pm \sqrt{100}}{6} \\ &= \frac{2 \pm 10}{6} \end{aligned}$$

Therefore,

$$\begin{aligned} x &= \frac{2 - 10}{6} \\ &= \frac{-8}{6} \\ &= -\frac{4}{3} \end{aligned}$$

or

$$\begin{aligned} x &= \frac{2 + 10}{6} \\ &= \frac{12}{6} \\ &= 2 \end{aligned}$$

Therefore, the solutions are $x = -\frac{4}{3}$ and $x = 2$.

3. (a) Determine the nature of the roots of $x^2 + 2x + 5 = 0$. Using the discriminant

$$\Delta = b^2 - 4ac$$

with $a = 1$, $b = 2$ and $c = 5$, gives

$$\begin{aligned} \Delta &= 2^2 - 4(1)(5) \\ &= 4 - 20 \\ &= -16 \end{aligned}$$

Since $\Delta < 0$, the quadratic has no real roots, i.e. the curve does not cross the x-axis.

(b) Determine the nature of the roots of $x^2 - 6x + 9 = 0$. Using the discriminant

$$\Delta = b^2 - 4ac$$

with $a = 1$, $b = -6$ and $c = 9$, gives

$$\begin{aligned} \Delta &= (-6)^2 - 4(1)(9) \\ &= 36 - 36 \\ &= 0 \end{aligned}$$

Since $\Delta = 0$, the quadratic has a repeated root, i.e. the curve crosses the x-axis once.

Solving:

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{- -6 \pm \sqrt{(-6)^2 - 4(1)(9)}}{2(1)} \\ &= \frac{6 \pm \sqrt{0}}{2} \\ &= \frac{6}{2} \\ &= 3 \end{aligned}$$

Hence, $x = 3$. Therefore, the equation has a repeated root at $x = 3$.

(c) Determine the nature of the roots of $2x^2 + x - 4 = 0$. Using the discriminant

$$\Delta = b^2 - 4ac$$

with $a = 2$, $b = 1$ and $c = -4$, gives

$$\begin{aligned} \Delta &= 1^2 - 4(2)(-4) \\ &= 1 + 32 \\ &= 33 \end{aligned}$$

Since $\Delta > 0$, the quadratic has two real roots, i.e. the curve crosses the x-axis twice.

Solving:

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-1 \pm \sqrt{1^2 - 4(2)(-4)}}{2(2)} \\ &= \frac{-1 \pm \sqrt{33}}{4} \end{aligned}$$

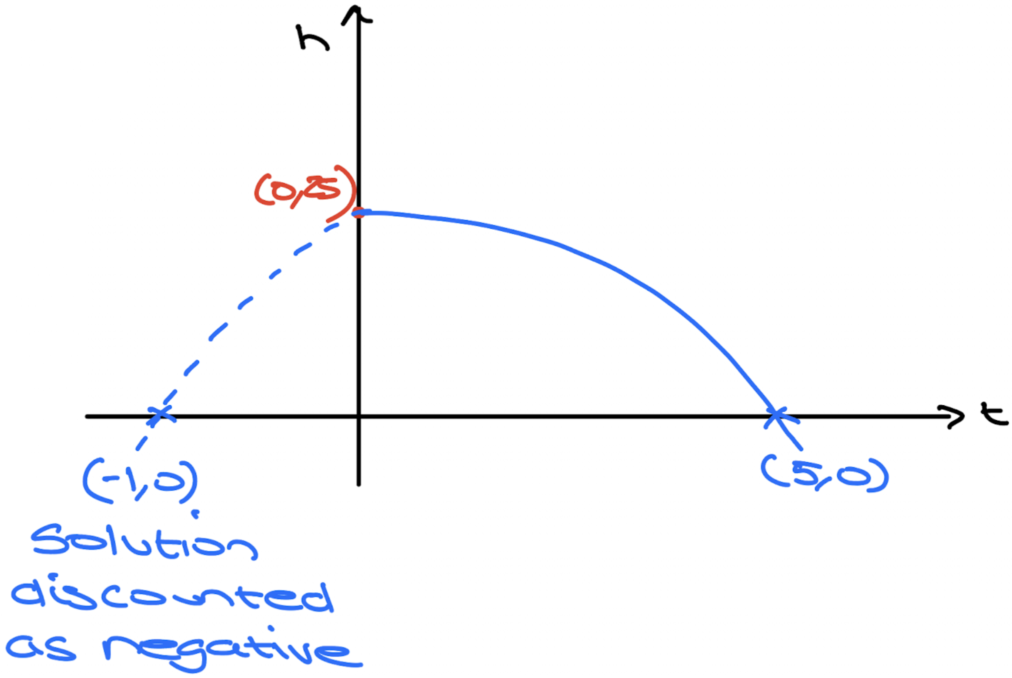
Hence, $x = -1.69$ and $x = 1.19$, both rounded to 2 decimal places.

4. (a) The ball reaches the ground at a height of zero metres, i.e $h = 0$. Therefore we wish to find t when $0 = -5t^2 + 20t + 25$. Using the quadratic formula gives:

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-20 \pm \sqrt{20^2 - 4(-5)(25)}}{2(-5)} \\ &= \frac{-20 \pm \sqrt{900}}{-10} \\ &= \frac{-20 \pm 30}{-10} \end{aligned}$$

Therefore the solutions are $t = 5$ and $t = -1$. In terms of the context of the question, a negative time can be discounted. Thus, the ball reaches the ground at $t = 5$ seconds.

(b) The sketh is below. Note that the y-intercept is at $(0, 25)$, which would be the initial height of the ball.


h
t
(0,25)
(-1,0)
(5,0)
Solution discounted as negative